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440C STAINLESS STEEL

Penn Stainless inventory now includes Alloy 440C (UNS S44000) in sheet, sheet coil, plate, round bar, processed flat bar and tubular products.

+ OVERVIEW

GENERAL PROPERTIES

Stainless steels are high-alloy steels which have high corrosion resistance compared to other steels due to the presence of large amounts of chromium. Based on their crystalline structure, they are divided into three types such as ferritic, austenitic, and martensitic steels. Another group of stainless steels are precipitation-hardened steels. They are a combination of martensitic and austenitic steels.

Grade 440C stainless steel is a high carbon martensitic stainless steel. It has high strength, moderate corrosion resistance, and good hardness and wear resistance. Grade 440C is capable of attaining, after heat treatment, the highest strength, hardness and wear resistance of all the stainless alloys. Its very high carbon content is responsible for these characteristics, which make 440C particularly suited to such applications as ball bearings and valve parts.

APPLICATIONS

Typical applications include:

Rolling element bearings

Valve seats

High quality knife blades

Surgical instruments

Chisels

SPECIFICATIONS

UNS: ASTM/ASTE S44000

DIN: 1.4125

+ RESISTANCE

CORROSION RESISTANCE

Good resistance to the atmosphere, fresh water, foods, alkalies and mild acids. Best resistance in the hardened and tempered and passivated condition. A smooth polished surface also assists.

The corrosion resistance of grade 440C approximates that of grade 304 in many environments.

HEAT RESISTANCE

Not recommended for use in temperatures above the relevant tempering temperature, because of reduction in mechanical properties by over-tempering.

+ PROCESSING

HEAT TREATMENT

Annealing – Full anneal – 850-900°C, slow furnace cool to about 600°C and then air cool. Sub-critical Annealing – 735-785°C and slow furnace cool.

Hardening – Heat to 1010-1065°C, followed by quenching in warm oil or air. Oil quenching is necessary for heavy sections. Immediately temper at 150-370°C to obtain a wide variety of hardness values and mechanical properties as indicated in the accompanying table.

Tempering in the range 425-565°C is to be avoided because of reduced impact resistance and corrosion resistance. Tempering in the range 590-675°C results in lower hardness (the product become machinable) and high impact resistance.

WELDING

If welding is necessary pre-heat at 250°C and follow welding with a full anneal. Grade 420 filler will give a high hardness weld (although not as high as the 440C), but 309 or 310 will produce soft welds with higher ductility.

MACHINING

In the annealed condition this grade is relatively easily machined; approximately the same as for high speed steel. Chips are tough and stringy so chip breakers are important. If these grades are hardened machining becomes more difficult and probably impossible.

+ PROPERTIES

Composition

Grade		C	Mn	Si	P	S	Cr	Mo	Ni	N
440C	min.	0.95	-	-	-	-	16.00	-	-	-
	max.	1.20	1	1	0.04	0.030	18.00	0.75		

Mechanical Properties

Tempering Temperature (°C)	Tensile Strength (MPa)	Yield Strength 0.2% Proof (MPa)	Elongation (% in 50mm)	Hardness Rockwell (HR C)	Impact Charpy V (J)
Annealed*	758	448	14	269HB max#	-
204	2030	1900	4	59	9
260	1960	1830	4	57	9
316	1860	1740	4	56	9
371	1790	1660	4	56	9

* Annealed properties are typical for Condition A of ASTM A276# Brinell Hardness is ASTM A276 specified maximum for annealed 440A, B and C.

Physical Properties

Grade	Density (kg/m ³)	Elastic Modulus (GPa)	Mean Coefficient of Thermal Expansion (mm/m/°C)			Thermal Conductivity(W/m.K)		Specific Heat 0-100°C (J/kg.K)	Electrical Resistivity (nW.m)
			0-100°C	0-200°C	0-600°C	at 100°C	at 500°C		
440 C	7650	200	10.1	10.3	11.7	24.2	-	460	600

Possible Alternative Grades

Grade	Why it might be chosen instead of 440C
440A/B	Slightly softer and more corrosion resistant grade needed
440F	High machinability required, with same hardness and hardenability as 440C
420	Lower strength and hardness needed than any of the 440 grades
416	Higher machinability required, and the much lower hardness and strength is still adequate

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SUBSCRIBE

Our stainless inventory and processing centers are located in **Quakertown, PA and Jackson Center, PA.**

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